

AgroLink

Ethiopia's Smart Agricultural Management System

Bridging farmers, agents, regional administrators, and buyers through digital innovation to transform agricultural trade in Ethiopia.



Project Overview

AgroLink is a comprehensive digital agricultural platform designed specifically for Ethiopian agricultural stakeholders. It creates structured connections between farmers, agents, regional administrators, and buyers—enabling verified product trading, transparent pricing, and reliable payments.

For Low Connectivity

Works in areas with limited internet access through smart caching and voice-based communication.

Trust-Based Trade

Includes product verification and secure payment through local systems like Chapa.

Local Payment Integration

Seamlessly integrates with Ethiopian payment infrastructure for verified transactions.



The Challenge

→ Disconnected Systems

Farmers lack structured channels to reach verified buyers, leading to wasted products and missed income opportunities.

→ Unfair Pricing

Without market transparency, intermediaries exploit farmers through information asymmetry and arbitrary price-setting.

→ Payment Risks

Manual cash transactions lack verification, creating fraud risks and payment disputes that erode trust.

→ Communication Gaps

Farmers receive no timely alerts about price changes, market demand, or critical weather-related disasters.

System Roles & Responsibilities

Superadmin

Oversees the entire system. Creates and manages regional administrators and agents, ensuring consistent platform governance and data integrity across all woreda-level operations.

Admin/Agent

Operates at woreda or kebele level. Registers farmers, records product details, verifies quality, and posts approved items to the buyer marketplace for transaction processing.

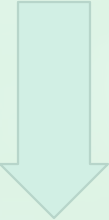
Farmer

Delivers products to the nearest agent or admin office. Receives voice notifications in their preferred language about successful sales, price updates, and critical disaster alerts.

Buyer

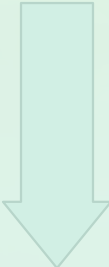
Accesses the marketplace dashboard to browse verified agricultural products. Purchases items and completes payment securely through the integrated Chapa payment gateway.

System Workflow



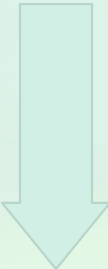
Registration

Superadmin creates regional admins and agents; admins register farmers with contact and banking details.



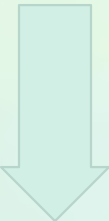
Product Collection

Farmers deliver products to nearby agents, who record details including type, quantity, quality grade, and harvesting date.



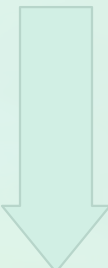
Verification & Publishing

Admins verify product quality and authenticity, then post approved items with pricing to the centralized buyer marketplace.



Purchase & Payment

Buyers browse listings, select products, and complete transactions through Chapa's secure payment gateway.



Notifications

Farmers receive voice alerts in Amharic, Afan Oromo, or Tigrigna about sales, price changes, and weather emergencies.

Core Features



Multi-Role Access

Customized dashboards for superadmins, regional admins, agents, farmers, and buyers with role-specific functionality.



Farmer & Product Registry

Centralized database capturing farmer identities, banking information, and detailed product records for accountability.



Chapa Payment Integration

Secure, verified transactions through Ethiopia's leading payment gateway, ensuring funds reach farmers reliably.



Voice Notification System

Automated voice alerts in local languages notify farmers of sales, market prices, and critical disaster warnings in real-time.



Buyer Marketplace

Intuitive interface for buyers to browse verified agricultural products, compare prices, and place orders conveniently.



Future Call Center

Planned direct farmer support line providing personalized assistance for inquiries and technical issues.



Technical Architecture

Frontend

React.js with Tailwind CSS delivers responsive, accessible user interfaces across devices and screen sizes.

Backend

Node.js and Express.js provide scalable server infrastructure for handling concurrent requests and data processing.

Database

MongoDB with Prisma ORM ensures flexible data modeling and reliable data persistence across the platform.

Authentication

JWT-based login system secures user access and maintains session integrity across all platform roles.

Integrations

- Chapa API for payment processing and transaction verification
- gTTS (Google Text-to-Speech) for multilingual voice notifications



Built for Ethiopia's Reality

Authentic Agricultural Flow

Mirrors the actual Ethiopian supply chain—local agents collect, admins verify, buyers access—ensuring system adoption among stakeholders familiar with existing trade patterns.

Low-Connectivity Design

Optimized for areas with intermittent internet through offline-capable features and efficient data synchronization when connectivity resumes.

Voice-First Communication

Recognizes literacy challenges by prioritizing voice alerts in Amharic, Afan Oromo, and Tigrigna—ensuring all farmers receive critical information.

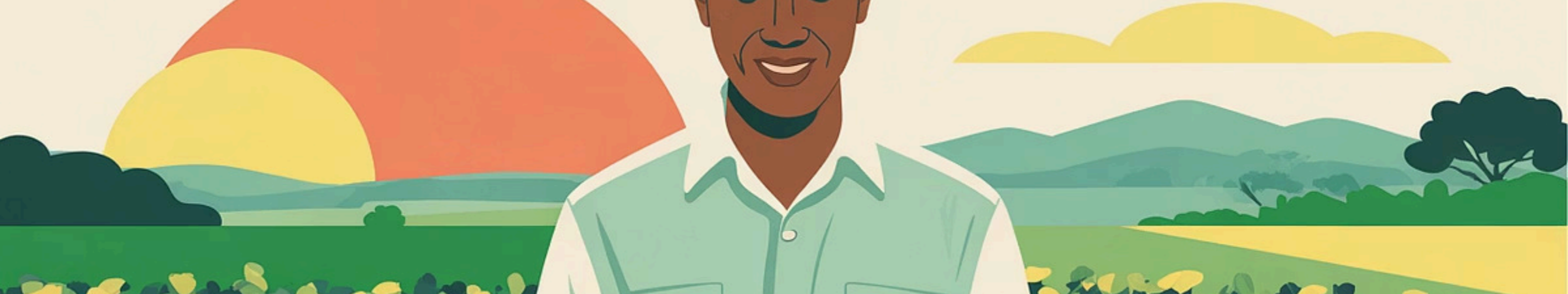
Local Payment Infrastructure

Integrates with Chapa, leveraging existing Ethiopian digital payment ecosystems farmers and buyers already understand and trust.

Impact & Achievements

Direct Market Access	Transparent Pricing	Verified Transactions
Real-Time Alerts	Digital Inclusion	

AgroLink empowers Ethiopian farmers by eliminating intermediary exploitation, enabling direct connections to verified buyers, and delivering timely market intelligence and disaster warnings. The platform strengthens Ethiopia's Digital Agriculture strategy while fostering inclusive economic growth in rural communities.



The Future of AgroLink

Planned Enhancements

- Direct call center support for farmer assistance
- AI-powered crop advisory and market forecasting
- Mobile offline-first application for unreliable connectivity
- SMS backup notifications for critical alerts

Vision

AgroLink represents a realistic, locally-tailored digital solution that transforms Ethiopian agriculture through digital trust, verified trade, and farmer empowerment. By bridging information gaps and enabling secure transactions, the platform catalyzes sustainable agricultural growth across Ethiopia's woreda and regional economies.